Natural Language Processing

Exercise – Summer term 2026

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Organization

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Programming Project: **Forensic Linguist Agent**

- Develop a modular agent system for forensic authorship analysis.
- Focus on *idiolectal style*: how a text is written rather than what it is about.
- A coordinator agent selects and executes reusable linguistic analysis skills.
- Ensure a transparent and reproducible workflow with inspectable intermediate results and uncertainty.
- Evaluate the system on authorship datasets and compare agent-based workflows to simpler baselines.

Agentic Workflow

What is an agent?

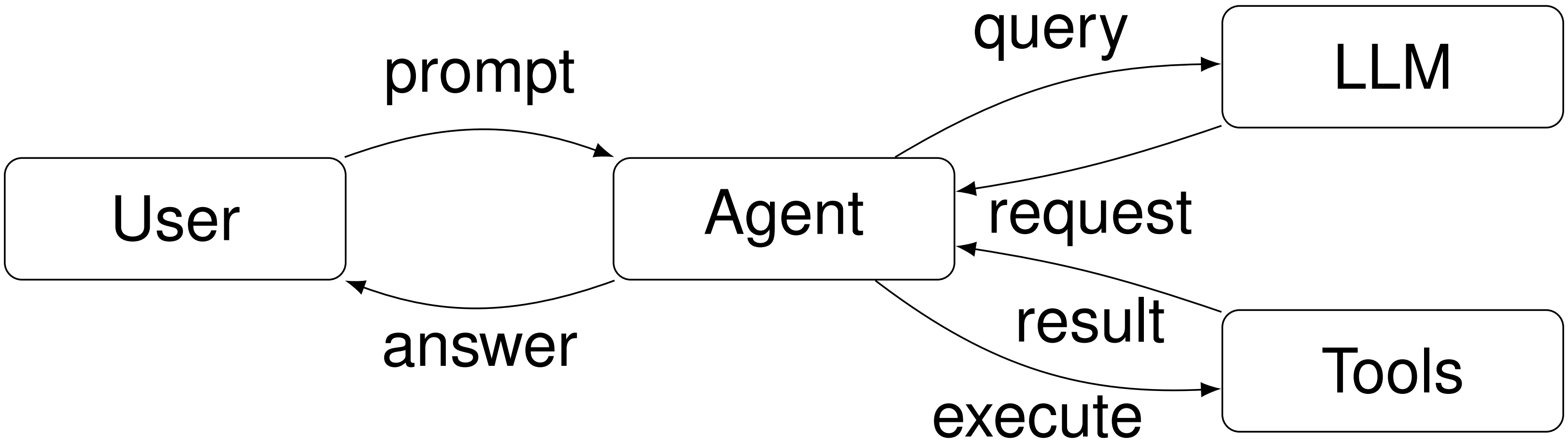

Figure: Simplified agentic workflow. A user prompt is processed by an agent that queries an LLM. The LLM may request tool calls, which are executed by the agent and returned as results. This loop continues until the agent produces a final answer.

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Details

Reminder of 28.04.2026

Your tasks include:

- Implement **tools** for linguistic analysis (e.g., linguistic feature extraction, stylometric analysis).
- **Persist** imediate **results** and **uncertainty** in a transparent way.
- Connect with a backend LLM.
- Identify risks and limitations of the agentic workflow and implement mitigation strategies.
- Evaluate the agent on benchmark datasets and compare to non-agentic baselines.
- Write an **4-page report** summarizing your approach, results, and insights in a two-column format including figures, tables, and references.

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Report

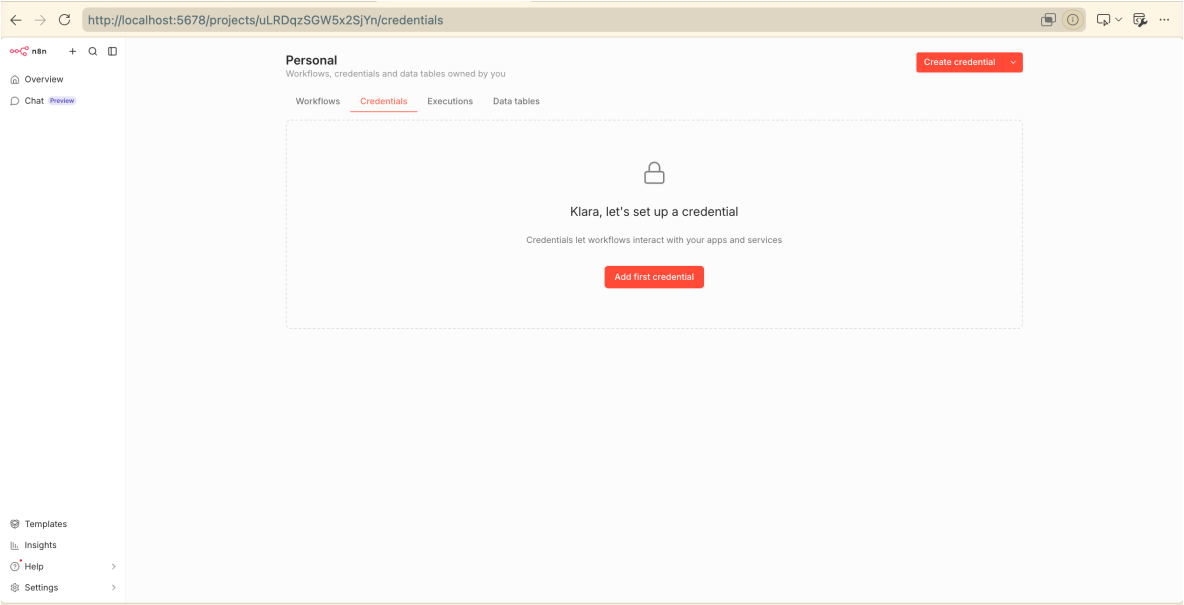
Checklist for the progress meeting:

- ❏ LaTeX ACM **template**
- ❏ 4-pages in two-column format including figures, tables, and references
- ❏ content [webis checklist]:
  - Title, author(s), abstract (5–10 sentences), keywords
  - Introduction
  - Related Work
  - Approach
  - Experiments
  - Conclusion
  - References
- ❏ Evaluate preliminary results and identify next steps.

n8n Setup I

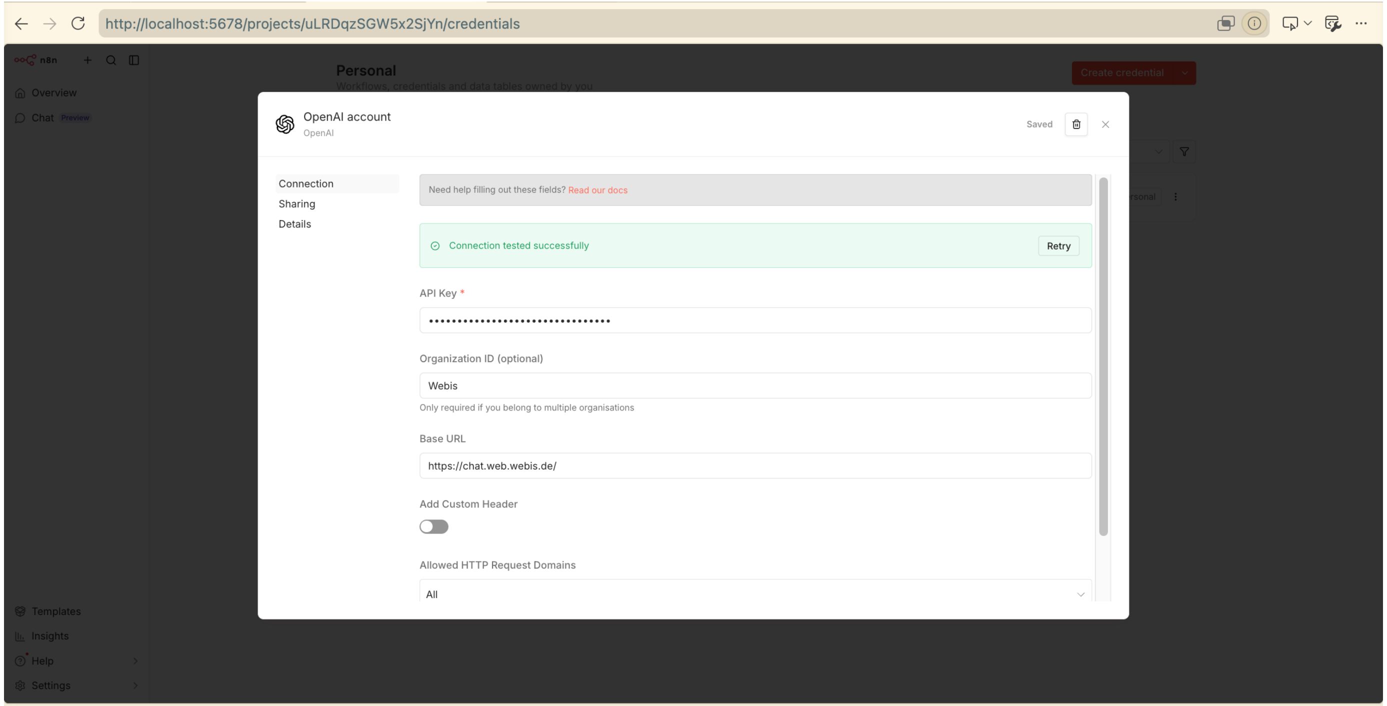
Start n8n

The *Docker Compose* configuration for n8n and Redis is available in the repository.
To start both services, run `docker compose up -d` from the repository root.
Once the containers are running, access the n8n editor at `http://localhost:5678`.



n8n Setup II

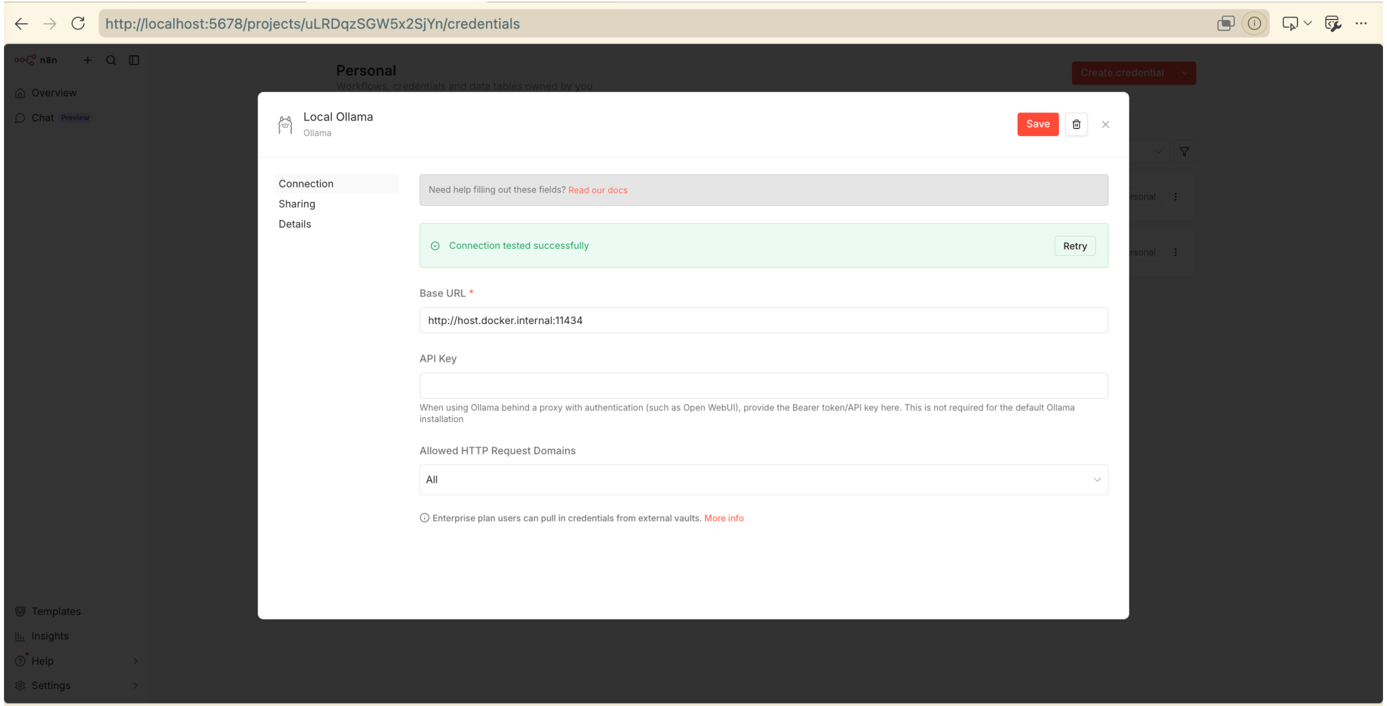
Configure the LLM Backend



Configure the Webis-hosted LLM using the OpenAI-compatible endpoint [n8n documentation].

n8n Setup III

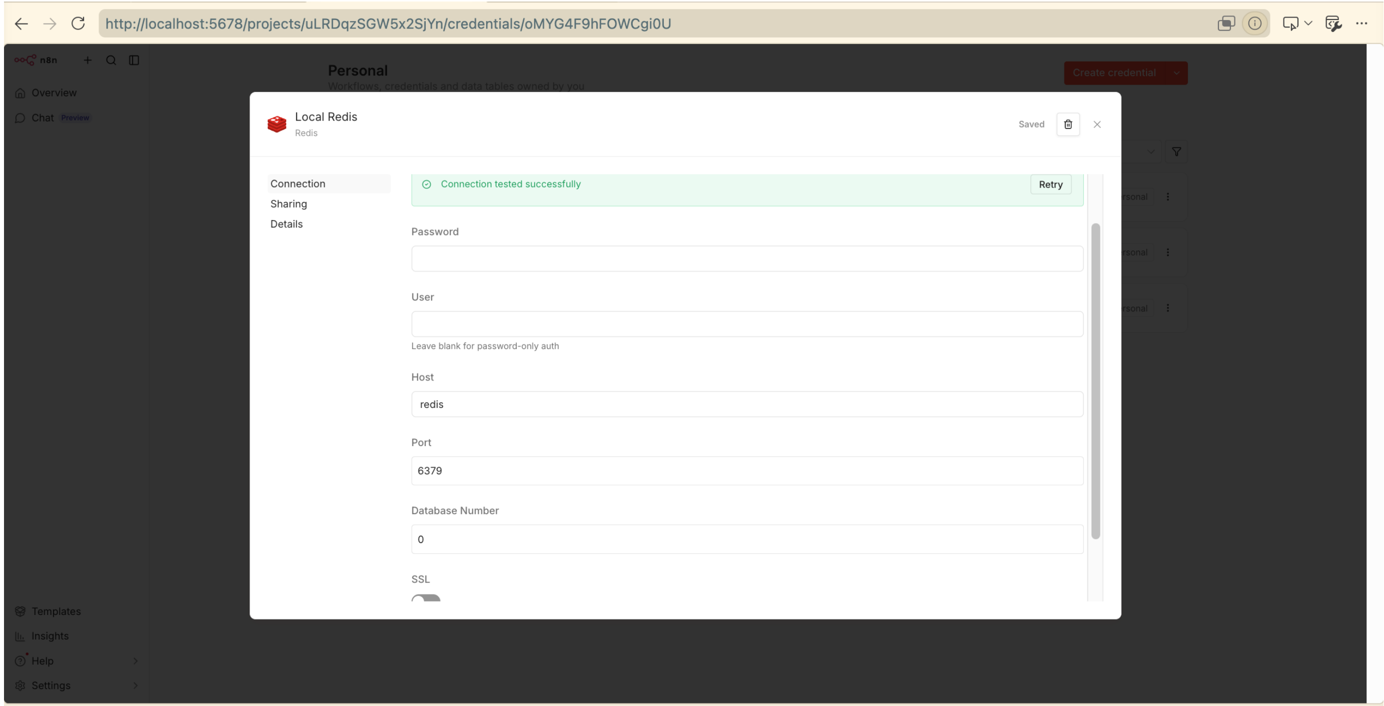
Configure the LLM Backend



Configure locally hosted LLM using the Ollama-compatible endpoint [n8n documentation].

n8n Setup IV

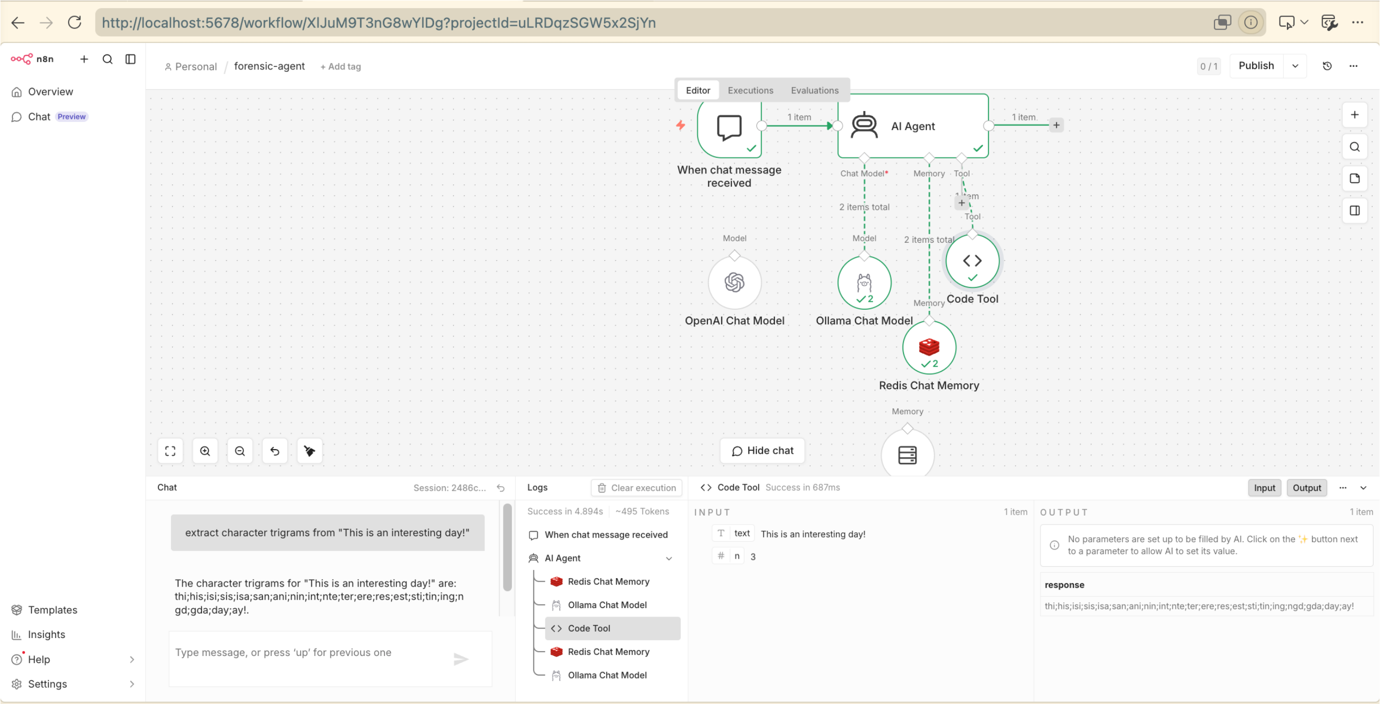
Configure the Redis Database



Configure the local Redis database. Unlike the setup described in the n8n documentation, a local Redis Docker container does not require authentication. Leave the username and password fields empty.

Persistent Storage

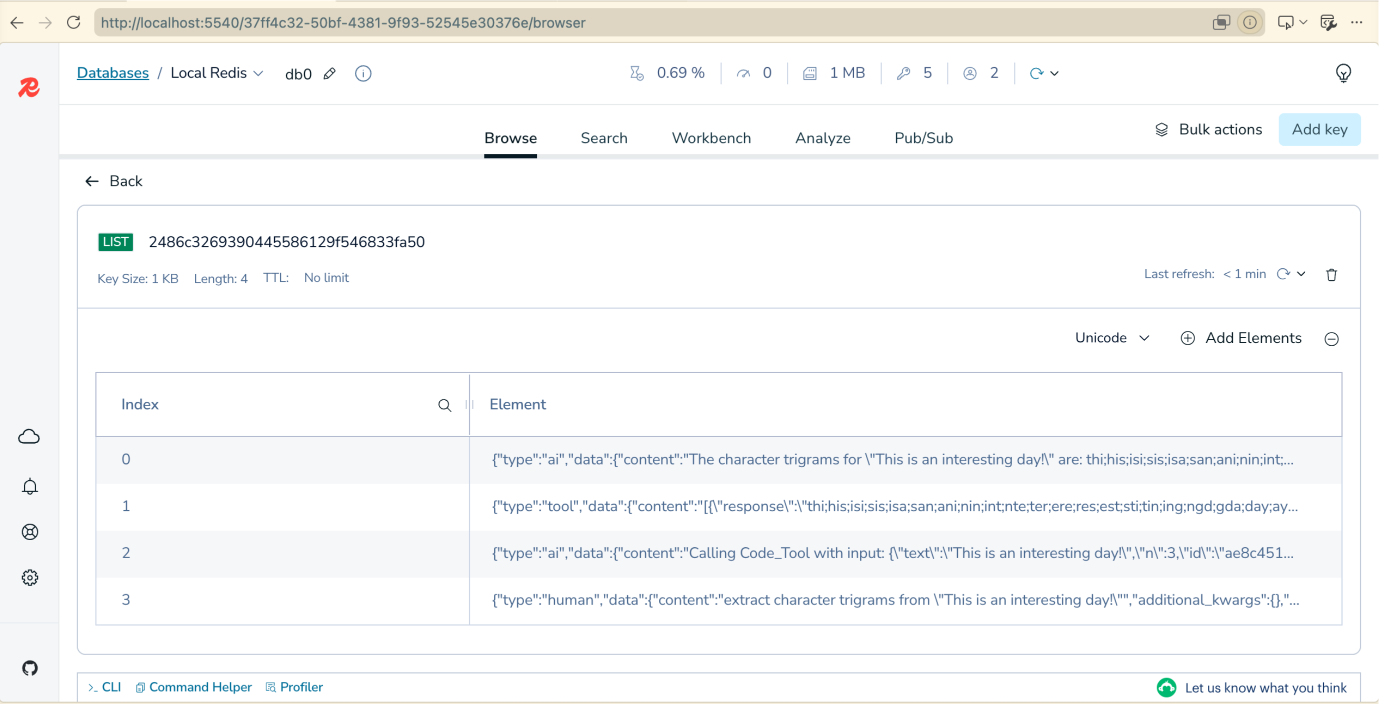
Redis Database



Working example of an n8n workflow using a local Ollama model (`granite4:3b`), a local Redis database, and a simple character n-gram extraction tool.

Persistent Storage

Redis Database



By mounting host directories (cf. repository), Redis data persists even after containers are stopped or recreated. The database can be explored using Redis Insight at `http://localhost:5540/`.

AI Agent

Configurations

System Prompt:

```
You are Klara’s local AI assistant.

For requests about n-grams:
1. Invoke the connected tool named ngram_tool and do not use memory.
2. Do not write “Calling ngram_tool”.
3. Do not describe the tool call.
4. After the tool result is returned, copy the tool result exactly.
5. Never calculate n-grams yourself.

If the user does not provide n, use n = 3.
```

- Require Intermediate Steps: `true`

Ollama Chat Model

Configurations

- Sampling Temperature: `0,1`

Code Tool

Configurations

- **Description:** `Call this tool to obtain character n-grams separated by ";".`
- **Language:** `JavaScript`

Code:

```javascript
const text = query.text ?? "";
const n = Number(query.n ?? 3);
const normalized = text.toLowerCase().replace(/\s+/g, "");
const grams = [];
for (let i = 0; i <= normalized.length - n; i++) {
grams.push(normalized.slice(i, i + n));
}
return grams.join(";");
```

Input Schema:

```json
{
"type": "object",
"properties": {
   "text": {
     "type": "string",
     "description": "Text to split into n-grams"
   },
   "n": {
     "type": "integer",
     "description": "Size of n-gram character sequences",
     "minimum": 2,
     "default": 3
   }
},
"required": ["text"]
}
```